Predictive and Generative Modeling of Therapeutic Antibody Developability Using Fine-Tuned Protein Language Models

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BMI-702 Final Presentation

Background and Motivation

- Antibodies are built from two chains:
  - **VH (Heavy-chain)**: Contributes to antigen binding
  - **VL (Light-chain)**: Pairs with VH, forms complete binding site

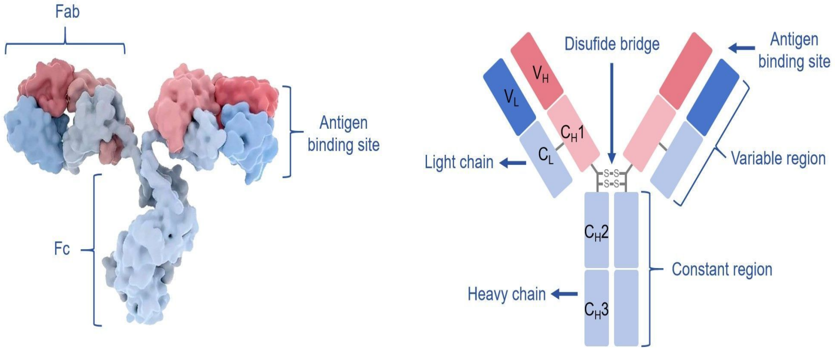
- Therapeutic antibodies are widely used, but expensive and slow to develop

  ↳ Candidates often fail due to poor developability

  - Developability metrics include biophysical properties:
    - Thermostability (T_m)
    - Self-association propensity (AC-SINS)
    - Hydrophobicity (HIC, SAP)
    - Solubility

- Lead optimization focuses on modifying antibody sequences to improve these properties

  ↳ Requires balancing developability while preserving binding function



Source: Creative Biolabs Antibody Development

Research Question: From Sequence to Developability

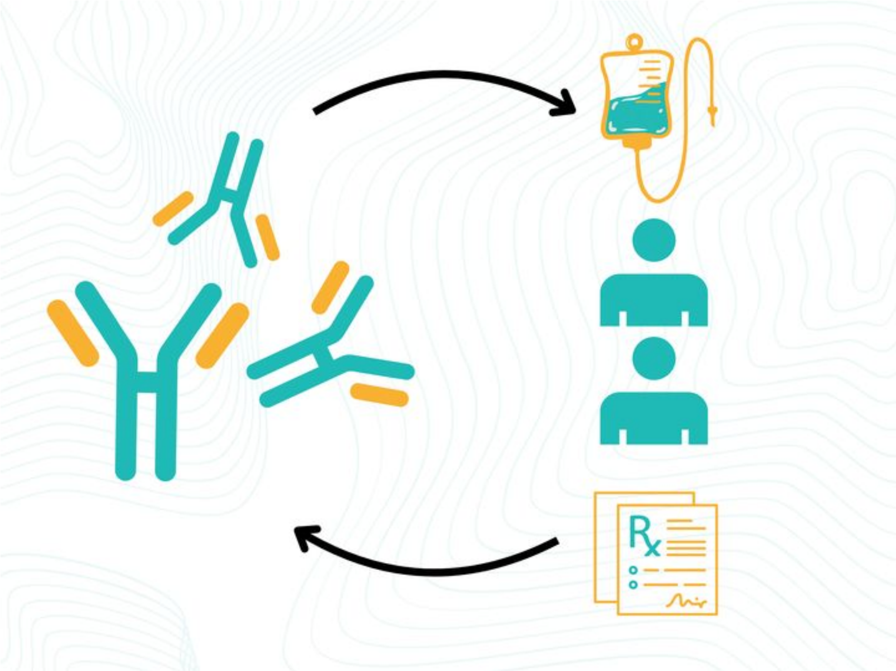
Main Research Question:

Can sequence-based protein language models learn representations that:

1. Predict developability properties and therapeutic approval?
2. Enable optimization of true developability properties through guided sequence generation?

- **Arsiwala et al. 2025**: Used experimentally derived features to predict developability and approval

- However, experimental features are not available early ↳ Motivates need for sequence-only ML models to predict developability properties



Source: Georgia Channing, Lood van Niekerk

Project Overview: Predict and Optimize Antibody Developability

Task A: Antibody Sequence Scoring

Input: VL and VH amino acid sequences

Output: Developability metrics and binary approval likelihood

Baseline Model: Ridge regression using p-IgGen for developability metrics, XGBoost for approval likelihood

Model: Frozen AbLang2 + Ridge; LoRA fine-tuned AbLang2 (r = 8, 0.82% trainable params) with MLP head

Training Data: Ginkgo Bioworks PROPHET-Ab (GDPa1_v1.2), 246 therapeutic antibodies

Task B: Antibody Sequence Generation

Input: Lead antibody CDR3, framework embedding, desired HIC and AC-SINS target scores, target antigen

Output: Generated CDR3 variants of the same length as the parent, with improved predicted developability (lower HIC, lower AC-SINS), predicted binding stability and affinity

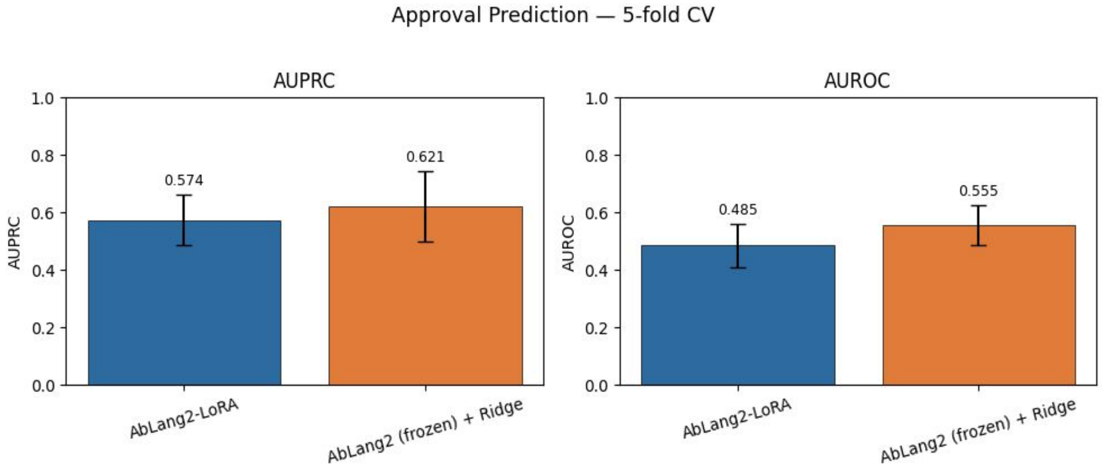
Model: Conditional flow matching in AbLang2 CDR3 PLS-reduced embedding space (480d → 10d), with FiLM conditioning on masked framework embeddings; AbLang2 MLM used as token-level sequence decoder, steered by flow model targets, cross attention on full antibody and antigen to predict binding affinity

Training Data: Ginkgo Bioworks PROPHET-Ab (GDPa1_v1.2), 197 therapeutic antibodies (folds 2–5), AbRank

Developability Metric Baseline Prediction (Ridge + p-IgGen)

Developability Target	GDPa1 Antibodies w/ Developability Target	Random-split Spearman ρ	CV Spearman ρ
HAC	94	0.329	0.621
PR_OVA	197	0.598	0.549
PR_CHO	197	0.438	0.496
AC-SINS_pH7.4	242	0.456	0.394
HIC	242	0.408	0.335
Titer	239	0.477	0.239
SMAC	242	0.250	0.207
Purity	242	-0.019	0.180
Tm2	193	-0.111	-0.084
SEC %Monomer	242	0.174	-0.101

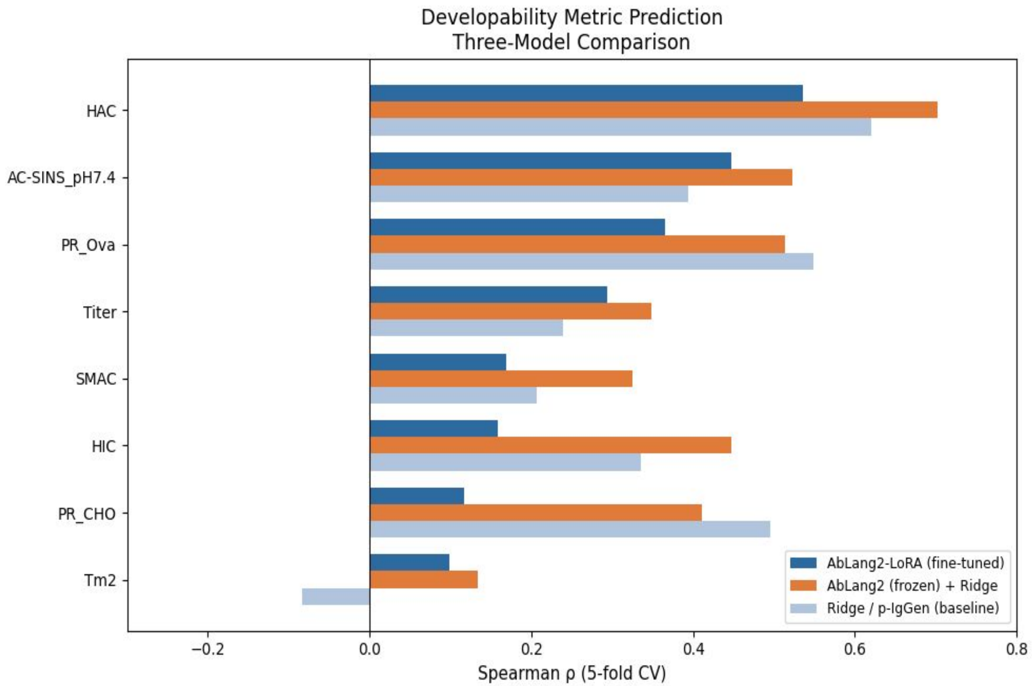
Antibody Approval Prediction: AbLang2 Ridge vs. AbLang2 LoRA



Model	AUPRC Mean	AUPRC Std. Dev	AUROC Mean	AUROC Std. Dev
AbLang2-LoRA	0.574	0.089	0.485	0.076
AbLang2 (frozen) + Ridge	0.621	0.123	0.555	0.070

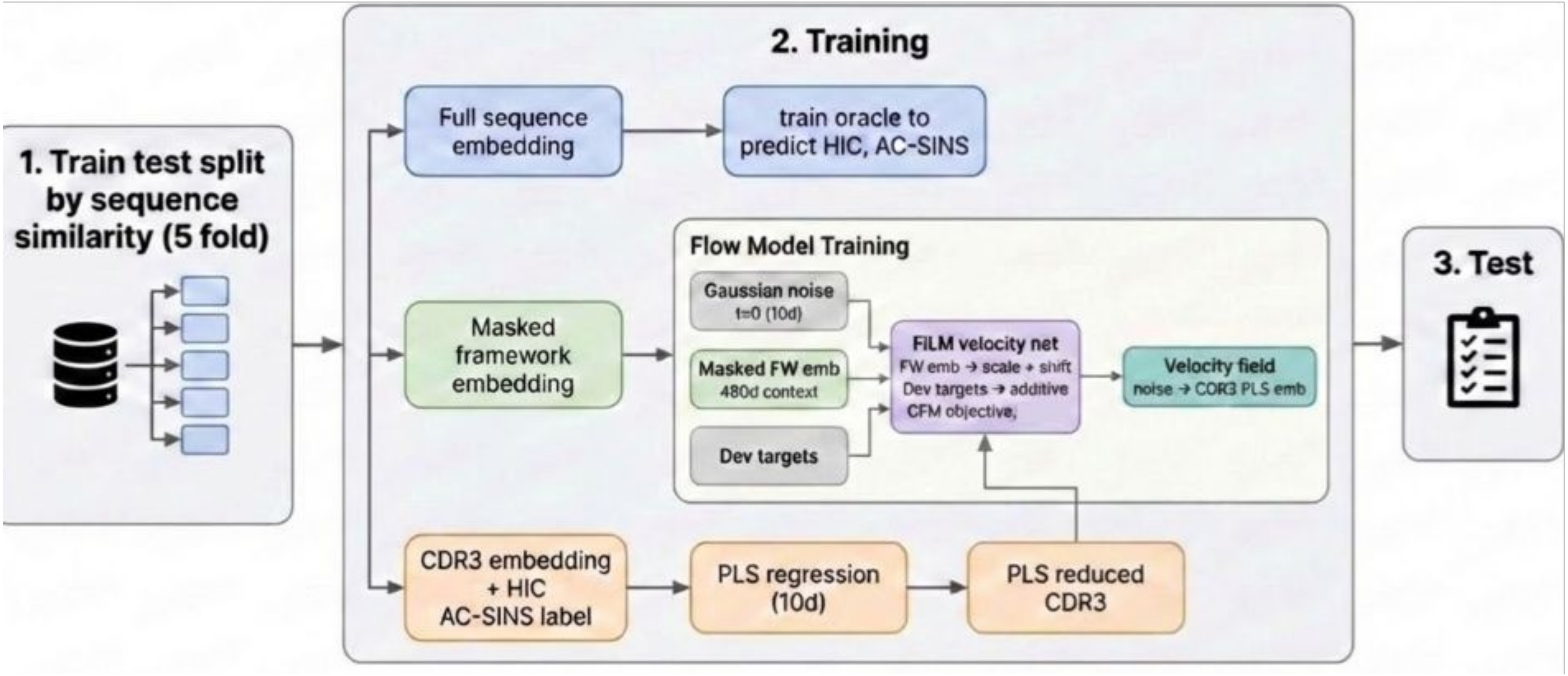
IgG Fold	AUPRC	AUROC	Loss at Epoch 30
0	0.622	0.519	0.658
1	0.655	0.540	0.644
2	0.456	0.424	0.618
3	0.502	0.385	0.608
4	0.634	0.558	0.639

Developability Metric Prediction: AbLang2 + Ridge vs. AbLang2 + LoRA vs. Ridge + p-IgGen



Developability Target	**AbLang2 + LoRA ρ**	**AbLang2 (frozen) + Ridge ρ**	**CV Ridge + p-IgGen ρ**
HAC	0.536	0.703	0.621
AC-SINS_pH7.4	0.448	0.523	0.394
PR_OVA	0.366	0.514	0.549
Titer	0.294	0.348	0.239
SMAC	0.169	0.325	0.207
HIC	0.158	0.447	0.335
PR_CHO	0.117	0.411	0.496
Tm2	0.099	0.134	-0.084

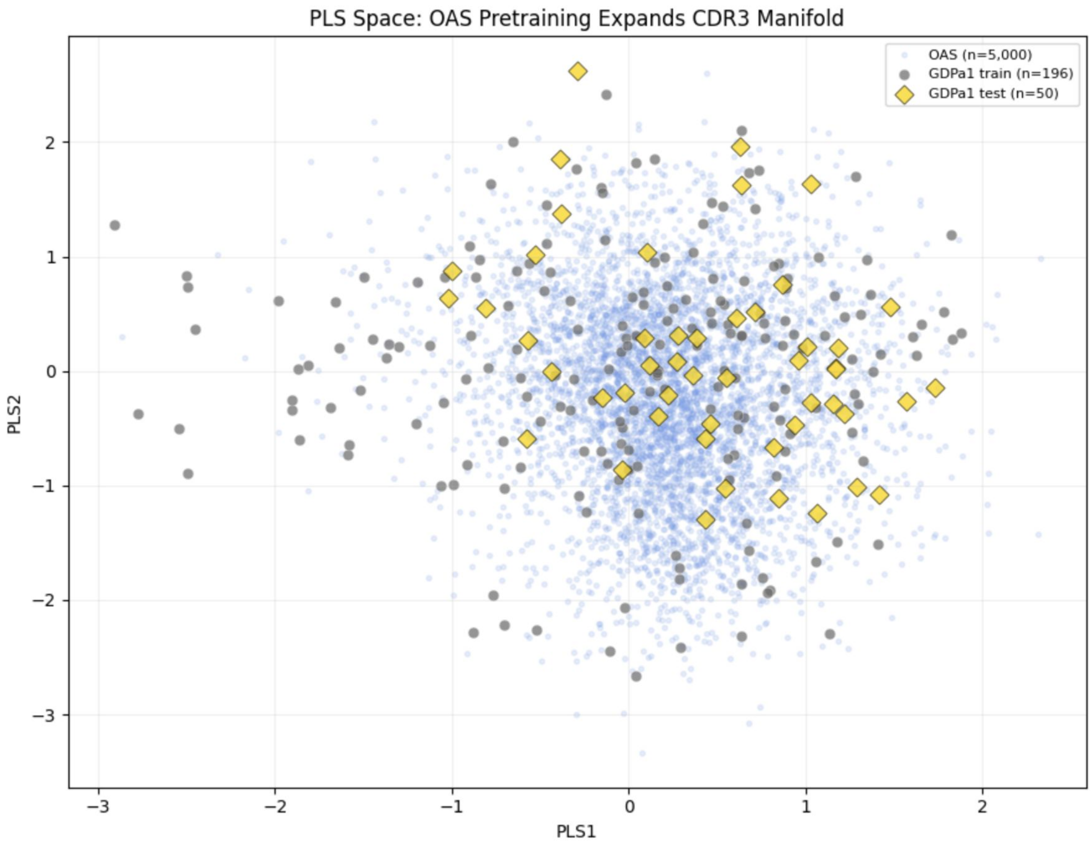
Generative Model Training Pipeline: Developability



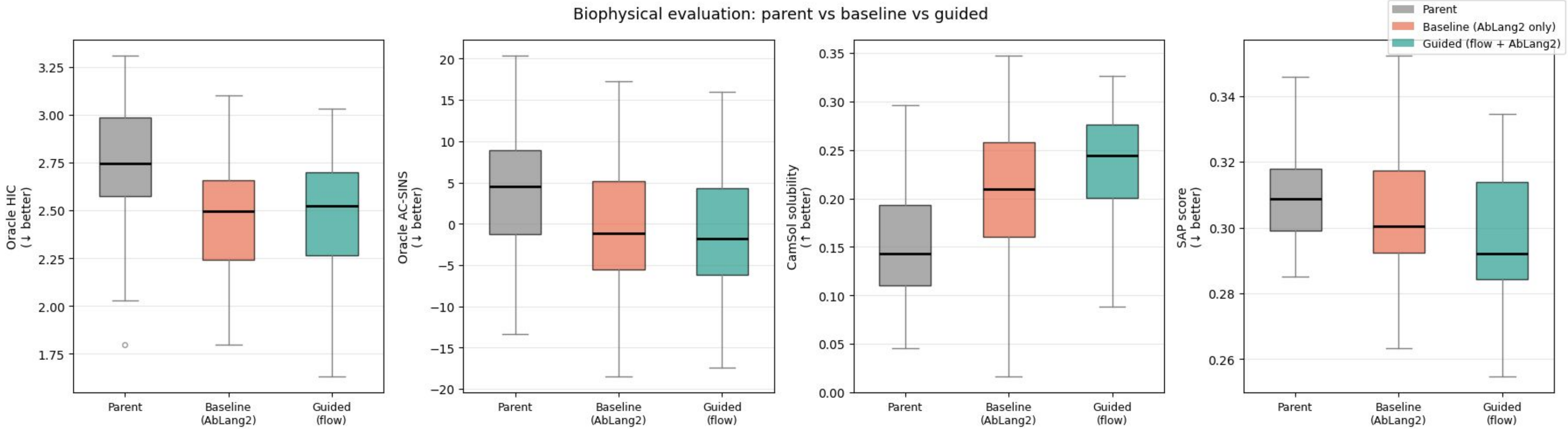
PLS: maximise covariance with HIC and AC-SINS

AC-SINS: Self-association
HIC: Hydrophobicity

OAS Pretraining: HCDR3 Projection in PLS Space



Pretraining Flow Model with OAS Sequences



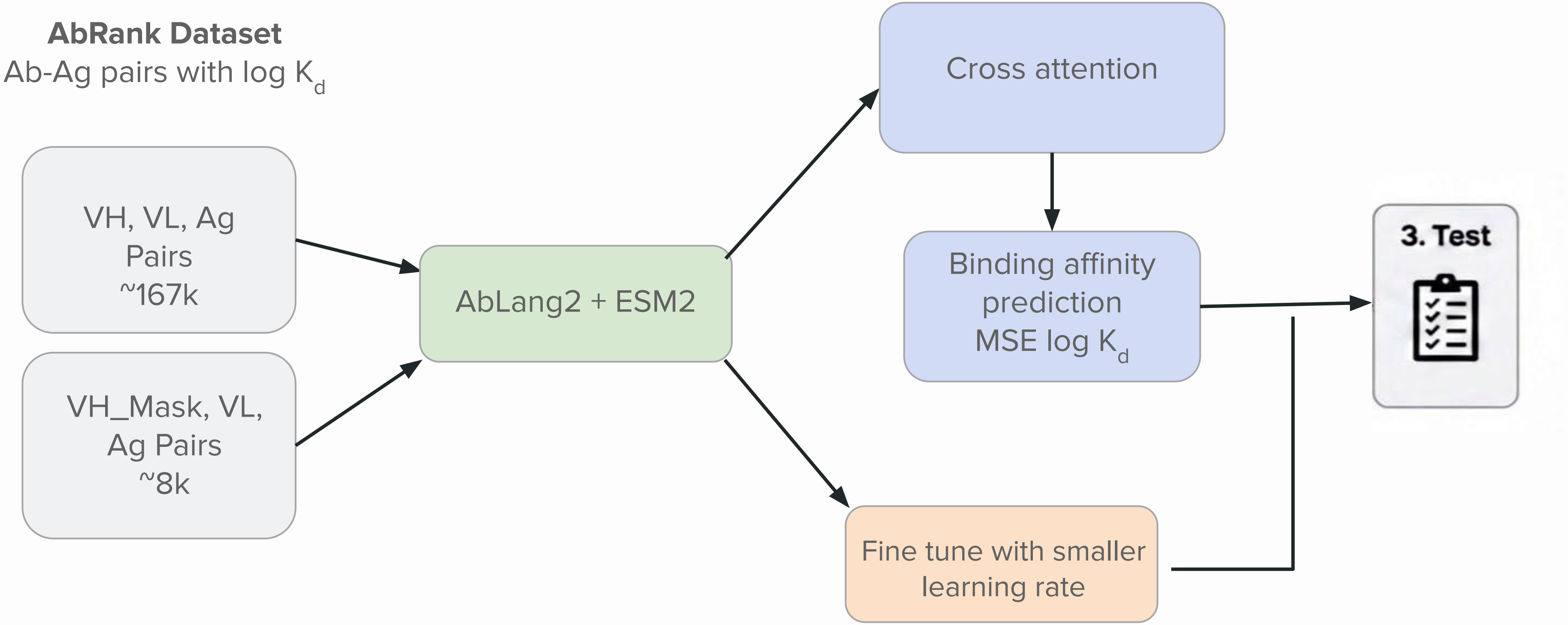
Metric	Parent	Baseline	Guided
Oracle HIC (↓)	+2.738 ± 0.048	+2.466 ± 0.043	+2.478 ± 0.047
Oracle AC-SINS (↓)	+3.741 ± 1.215	-0.545 ± 1.178	-0.696 ± 1.073
CamSol (↑)	+0.150 ± 0.008	+0.207 ± 0.010	+0.233 ± 0.009
SAP (↓)	+0.310 ± 0.002	+0.303 ± 0.003	+0.297 ± 0.003

Paired t-tests vs parent (* = p < 0.05):

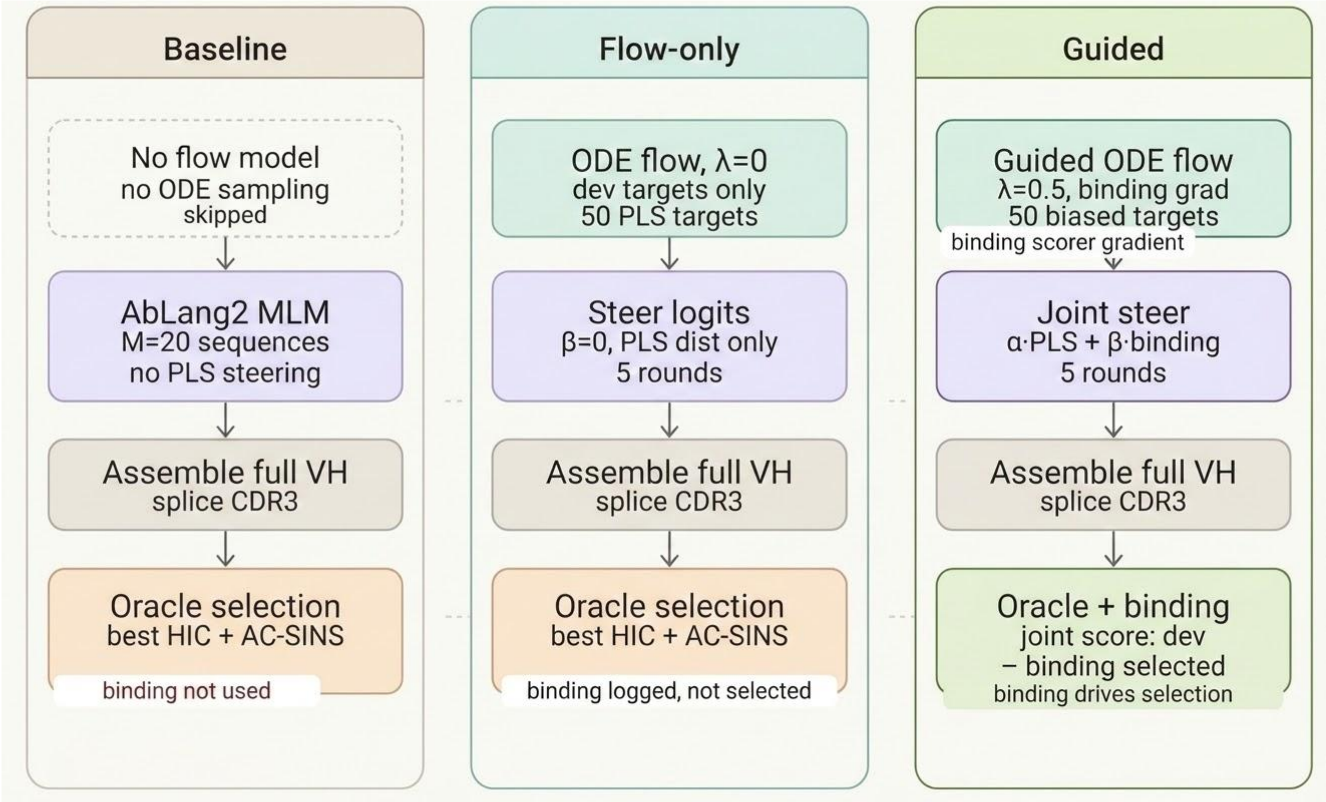
Metric	Baseline Δ / p	Guided Δ / p
Oracle HIC	Δ =-0.271 p=0.000*	Δ =-0.260 p=0.000*
Oracle SINS	Δ =-4.286 p=0.000*	Δ =-4.437 p=0.000*
CamSol	Δ =+0.058 p=0.000*	Δ =+0.084 p=0.000*
SAP	Δ =-0.007 p=0.007*	Δ =-0.013 p=0.000*

AC-SINS: Self-association — HIC: Hydrophobicity
CamSol: Computed solubility — SAP: Aggregation due to hydrophobic patches

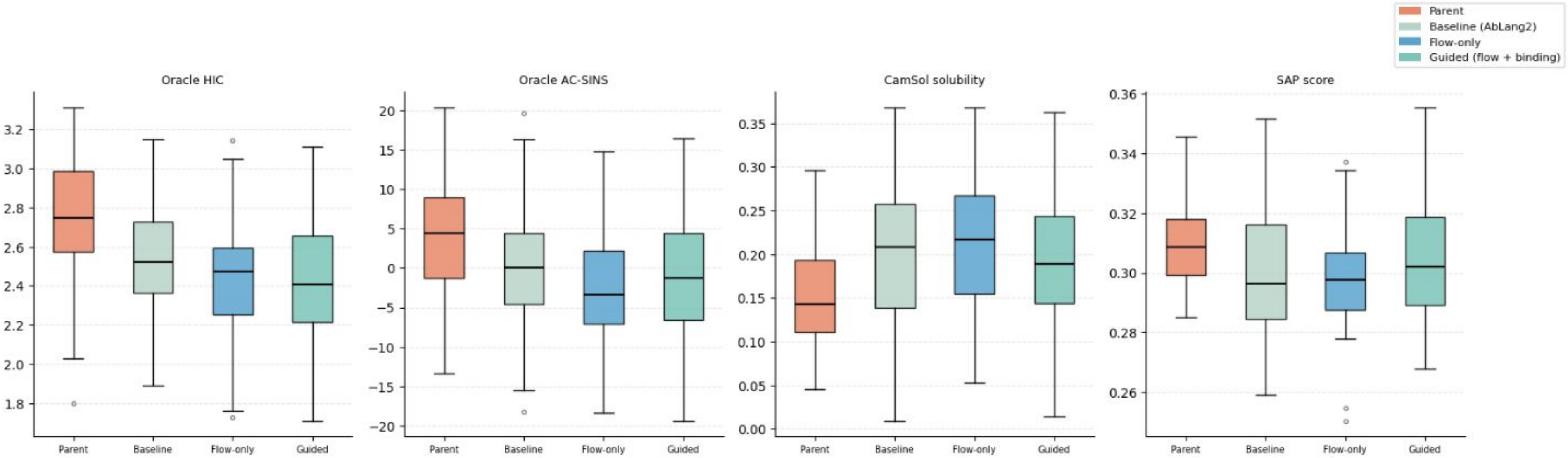
Generative Model Training Pipeline: Binding Affinity



Generation: Baseline, Flow only, Guided (developability + binding)



Flow Steering Improves Developability, Binding Scorer Perfoms Poorly



Metric	Dir	Baseline	Flow-only	Guided
Oracle HIC	↓	Δ =−0.203 ± 0.032 p=0.000*	Δ =−0.305 ± 0.032 p=0.000*	Δ =−0.326 ± 0.029 p=0.000*
Oracle AC-SINS	↓	Δ =−3.937 ± 0.584 p=0.000*	Δ =−6.022 ± 0.631 p=0.000*	Δ =−5.372 ± 0.602 p=0.000*
CamSol (↑ better)	↑	Δ =+0.051 ± 0.008 p=0.000*	Δ =+0.067 ± 0.009 p=0.000*	Δ =+0.044 ± 0.009 p=0.000*
SAP	↓	Δ =−0.011 ± 0.002 p=0.000*	Δ =−0.012 ± 0.002 p=0.000*	Δ =−0.006 ± 0.003 p=0.035*

* $p < 0.05$ (paired t-test vs parent)

AC-SINS: Self-association
CamSol: Computed solubility
HIC: Hydrophobicity
SAP: Aggregation due to hydrophobic patches

Key Takeaways

- **Limited measurement of binding affinity** leading to **poor performance** on **binding prediction** for unseen antigen antibody pairs

- **Limited performance gains from fine-tuning AbLang2 on GDPa1**
    - AbLang2 embeddings (frozen + Ridge) perform similarly to the p-IgGen baseline
    - LoRA fine-tuning overfits and suggests that sequence-based models capture only limited biophysical signal for developability prediction on the GDPa1 dataset

- **Conditional flow matching enables targeted HCDR3 optimization**
    - Generated variants show statistically significant improvements in predicted HIC, AC-SINS, CamSol solubility, and SAP relative to parent antibodies
    - Pretraining flow model’s conditional steering adds benefit over AbLang2's pretrained prior alone

Roles

Sidhant:

- Built baseline models for developability metrics (ridge regressions with p-IgGen) and antibody approval (XGBoost)
- Built regression oracles and LoRA fine tuned sequence based oracle (using AbLang2) for developability metric prediction and antibody approval prediction
- Pretrained and evaluated conditional flow model on OAS sequences and implemented parent vs. baseline vs. guided evaluations

Jack:

- Built generative pipeline for developability optimization with dimensionality reduction of CDR3 embeddings using PLS
- Trained a conditional flow matching model in reduced embedding space and implemented a steered sequence generation procedure using AbLang2 guided by flow model targets
- Binding affinity prediction pipeline